1) Known: $r = .001m$, $L = .05m$, $T_\infty = 25°C$, $h = 10\ W/m^2K$, $k = 200\ W/mK$

find: a) if base $T_b = 125°C$ find fin temp @ $x = L/2$ and $x = L$

b) fin effectiveness ε_f and fin efficiency η_f (look @ table 3.4)

c) fin efficiency η_f based on results from Table 3.5; compare result to (b) & discuss if Table 3.5 provides a good approximation to η_f

d) Thermal resistance of pin fin R_f

e) pin form 5x5 array mounted on base plate w/ side length $s = 0.03m$, find efficiency of fin array η_o & thermal resistance of fin array R_o. How much dissipation through unfined surface? How much through fin?

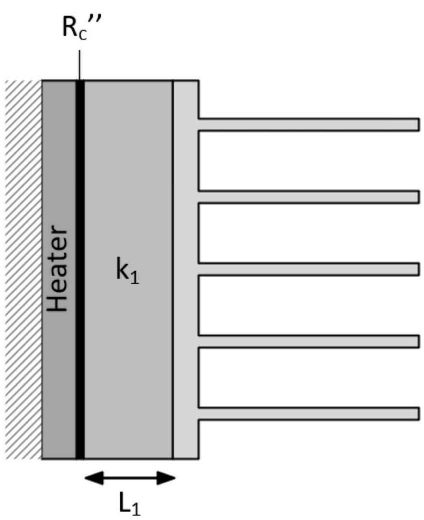
f) Add heat sink, $L_1 = 2cm$, $k_1 = 10W/mK$, $R_c = 10^{-4}\ m^2K/W$, $q'' = 1000W/m^2$ (dissipation from heat sink)

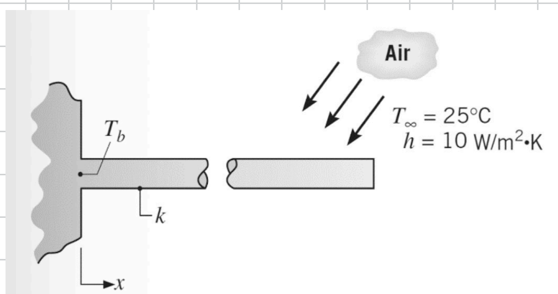
Draw thermal Resistance & calculate temp of thin film T_h, compare w/o heatsink how much temp reduction?

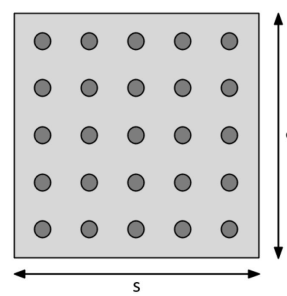
Assumptions:

1) Tip thermally insulated
2) Thermal insulation applied to back
3) Temp across fin uniform
4) no contact btw base & slab

Schematic:







Analysis:

a) $A_c = (\pi)(.001)^2 = 3.14\times10^{-6}\ m^2$

$P = 2\pi(.001) = 0.006283m$

$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(10)(.006283)}{(200)(3.14\times10^{-6})}} = 10m^{-1}$

$\frac{\theta}{\theta_b} = \frac{T - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L-x)}{\cosh mL} = \frac{\cosh 10(.25)}{\cosh(10\cdot(.5))} = 0.915$

$\frac{\theta}{\theta_b} = 91.5°C \rightarrow \theta = (.915°C)(T_b - T_\infty)$
$= (.915°C)(125°C - 25°C) = 91.5°C$

$\theta = T - T_\infty \rightarrow T = (25°C) + (91.5°C) = \boxed{116.5°C\ @\ x = L/2}$

$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL} = \frac{\cosh(0)}{\cosh(10\cdot.05)} = \frac{1}{1.1276} = 0.887$

$\theta = (100)(.887) = 88.7°C$ $\quad T = \theta + T_\infty = (88.7°C) + (25°C)$

$\boxed{T = 113.68°C\ @\ x = L}$

b) $\eta_f = \frac{q_f}{hPL\theta_b} = \frac{\sqrt{hPkA_c}\tanh(mL)}{hPL} = .924 = (92.4\%)$

$\varepsilon_f = \frac{M\tanh(mL)}{hA_c\theta_b} = \frac{\sqrt{hPkA_c}\,\cancel{\theta_b}\tanh(mL)}{hA_b\cancel{\theta_b}} = \sqrt{\frac{Pk}{hA_b}}\tanh(mL)$

$= \sqrt{\frac{(.006283)(200)}{(10)(3.14\times10^{-6})}}\tanh(.5)$ $\quad \boxed{\varepsilon_f = 92.45}$

c) $\eta_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(m(L + \frac{A_c}{P}))}{m(L + \frac{A_c}{P})} = \frac{\tanh 10(.05 + \frac{.002}{4})}{10(.05 + \frac{0.002}{4})}$

$\eta_f = 0.923 = \boxed{92.3\%}$

error: $\left(\frac{.924 - .923}{.924}\right)100 = \boxed{0.1\%\ \text{error so table 3.5 is a good approximation}}$

d) $q_f = \frac{\theta_b}{R_f} \rightarrow R_f = \frac{\theta_b}{q_f}$

$R_f = \frac{100}{.290} = \boxed{344.8\ K/W = R_f}$

e) $\eta_o = 1 - \frac{NA_f}{A_{tot}}(1-\eta_f) = 1 - \frac{25(.05)(.00628)}{(25(.5)(.00628) + (.03)^2) - 25(3.14\times10^{-6})} * (1-.924)$

$\eta_o = 0.931\ (100) = \boxed{93.1\%}$

$R_o = \frac{1}{\eta_o h A_{tot}} = \frac{1}{(.9318)(10)[A_{tot}]} = \boxed{12.38\ K/W}$

$q_{fins} = N q_f = (25)(.209) = \boxed{7.26W = q_{fin}}$

$q_{unfined} = h(A_b - A_{fin})\Delta T$

$= (10)(9\times10^{-4} - 25(3.14\times10^{-6}))(100)$

$\boxed{q_{unfined}} = 0.82\ W$

f)

with: $q'' = 1000\ W/m^2$

$q = \frac{q''}{A} = \frac{1000\ W/m^2}{(0.03)^2 m^2} = 1.11\times10^6\ W$

$R_{Total} = R_c + R_{slab} + R_f$

$= \frac{R_c''}{A} + \frac{L}{K_1 A} + \frac{1}{\eta_o h A_{tot}} = \left(\frac{10^{-4}}{9\times10^{-4}}\right) + \frac{.02}{(10)(9\times10^{-4})} + 12.4 = 14.73\ K/W$

$T_h - T_\infty = q R_{total}$

$T_h = T_\infty + q R_{tot} = 25°C + (.9)(14.73\ K/W) = \boxed{38.26°C}$

w/o: $R_{Total} = R_c + R_{slab} + R_{conv}$

$R_{tot} = \frac{R_c''}{A} + \frac{L}{K_1 A} + \frac{1}{hA} = (.111) + 2.22 + 111.1 = 113.442$

$T_h = T_\infty + q R_{Tot} = 25 + (.9)(113.442) = \boxed{127.1°C}$

With the heat sink there is an 88°C reduction

$\boxed{\Delta T = 88°C}$